Module 62.1: Credit Analysis for Corporate Issuers

SCHWESERNOTES - BOOK 3

LOS 62.a: Describe the qualitative and quantitative factors used to evaluate a corporate borrower's creditworthiness.

An analyst will use both qualitative and quantitative factors when evaluating the likelihood and impact of default by a corporate bond issuer.

Qualitative Factors

Qualitative factors include an issuer's business model, the degree of competition in its industry, its business risk, and the quality of its corporate governance.

Business model. A corporate issuer with high credit quality will have a business model with stable and predictable cash flows. For longer-term debt issues, a credit analyst should consider both the existing business model and any long-term changes to it that the issuer will need to make to remain competitive. Changes in the business model will increase business risk.

Industry competition. Less intensive competition is favorable for an issuer's credit quality. As with the business model, analysts need to consider any change in the competitive landscape over the long term.

Business risk. High-credit-quality issuers have low risk of unexpected deviations from expected revenues and margins. Business risk can originate from issuer-specific, industry-specific, or external sources.

Corporate governance. An issuer with high credit quality should have sufficient processes in place relating to the fair and legal treatment of debtholders. Specific concerns include debt covenants and accounting policies.

- *Covenants*. For unsecured investment-grade issuers, it is likely that covenants are primarily affirmative, relating to compliance with rules and laws, maintenance of company assets, and paying taxes. Analysts must assess the potential for management to issue additional debt that would dilute the claims of existing debtholders. High-yield issuers are likely to have negative covenants, restricting the issuer's ability to pay dividends or issue further debt, in addition to affirmative covenants. Here, analysts need to assess the past actions of management for evidence of preferential treatment of equity investors over debt investors, particularly if this led to credit rating downgrades (e.g., a debt-financed stock buyback program).
- *Accounting policies*. While evidence of fraud is an obvious concern, the use of aggressive accounting policies that accelerate revenue recognition, significant use of off-balance-sheet financing, a heavy preference for capitalizing spending rather than immediate expensing, and frequently changing auditors or the chief financial officer are also warning signs that the character of management may hurt the creditworthiness of the issuer.

Quantitative Factors

Quantitative modeling of the future performance of corporate issuers involves estimating future financial statements and cash flows of the issuer to identify key factors driving their probability of default and loss given default, and how these are likely to change over the economic cycle. While investors in unsecured investment-grade debt are likely to be primarily concerned about the probability of default increasing, investors in secured high-yield debt are also concerned about loss given default, due to the higher probability of default for high-yield debt.

Quantitative analysis can be performed on a top-down or bottom-up basis, or both. *Top-down* inputs relate to the macroeconomic cycle, the size of the industry and potential market share, and event risk related to potential external shocks. *Bottom-up* inputs relate to issuer-specific factors driving revenue, costs, balance sheet assets and liabilities, and future cash flows. A *hybrid* approach considers both top-down and bottom-up factors.

All else equal, companies are deemed to be of higher credit quality if the issuer has:

- Strong operating profits and recurring revenues
- Low levels of leverage and less reliance on debt in the capital structure
- High coverage of debt service payments with periodic income

- High levels of liquidity to meet short-term debt payments

LOS 62.b: Calculate and interpret financial ratios used in credit analysis.

Credit analysts calculate ratios to assess the creditworthiness of a company, to find trends over time, and to compare companies to industry averages and peers. Common ratios relate to profitability, coverage, and leverage.

Some key terms that are commonly used in calculating ratios are as follows:

- **Earnings before interest, taxes, depreciation, and amortization (EBITDA)**. EBITDA is a commonly used measure that is calculated as operating income plus depreciation and amortization. A drawback to using this measure for credit analysis is that it does not adjust for capital expenditures and changes in working capital, which are necessary uses of funds for a going concern. Cash needed for these uses is not available to debtholders.
- **Cash flow from operating activities (CFO)**. CFO is net cash paid or received in the continuing operations of the business. It is calculated as net income plus noncash charges minus increase in working capital. CFO is disclosed by companies in their cash flow statements.
- **Funds from operations (FFO)**. FFO is net income from continuing operations plus depreciation, amortization, deferred taxes, and noncash items. FFO is similar to CFO, except that FFO excludes changes in working capital.
- **Free cash flow (FCF)**. FCF is CFO minus fixed asset expenditures plus net interest expense. It represents the discretionary cash flow of a company because it could be paid to providers of financing after all the obligations of the company have been met.
- **Retained cash flow (RCF)**. RCF is operating cash flow minus dividends. Analysts may define operating cash as CFO, FFO, or another preferred measure.

Professor's Note

Retained cash flow is an analyst-based ratio rather than an accounting measure. It is defined as operating cash flow minus dividends. The analyst may choose a preferred definition of operating cash, with both CFO and FFO being popular measures.

Common credit analysis ratios are summarized in **Financial Ratios for Corporate Credit Analysis**.

Financial Ratios for Corporate Credit Analysis

Ratio Type	Ratio Name	Calculation	Indication of Higher Credit Quality
Profitability	EBIT margin	EBIT / revenue	Higher ratio
Coverage	EBIT to interest expense	EBIT / interest expense	Higher ratio
Leverage	Debt to EBITDA	Debt / EBITDA	Lower ratio
Leverage	RCF to net debt	RCF / (debt – cash and marketable securities)	Higher ratio

Example: Credit analysis based on ratios

An analyst is assessing the credit quality of York, Inc., and Zale, Inc. Selected financial information appears in the following table.

	York, Inc.	Zale, Inc.
Revenue	$2,200,000	$11,000,000
Depreciation and amortization	$220,000	$900,000
Earnings before interest and taxes	$550,000	$2,250,000
Cash flow from operations	$300,000	$850,000
Interest expense	$40,000	$160,000
Total debt	$1,900,000	$2,700,000
Cash and marketable securities	$500,000	$1,000,000
Dividends	$30,000	$200,000

Using profitability, coverage, and leverage ratios, explain how the analyst should evaluate the relative creditworthiness of York and Zale.

Answer:

Profitability ratios based on these data are as follows:

EBIT margin = EBIT / revenue:

York: $550,000 / $2,200,000 = 25%

Zale: $2,250,000 / $11,000,000 = 20.5%

York has higher EBIT margin, so from a profitability perspective, it has higher credit quality than Zale.

Coverage ratios based on these data are as follows:

EBIT / interest:

York: $550,000 / $40,000 = 13.8×

Zale: $2,250,000 / $160,000 = 14.1×

York has a slightly lower interest coverage ratio than Zale, suggesting slightly lower credit quality. The difference is small, though, and both companies can comfortably meet their interest payments from EBIT.

Leverage ratios based on these data are as follows:

Debt / EBITDA:

York: $1,900,000 / ($550,000 + $220,000) = 2.5×

Zale: $2,700,000 / ($2,250,000 + $900,000) = 0.9x

York is more leveraged than Zale, suggesting a lower credit quality for York.

RCF / net debt:

York: ($300,000 - $30,000) / ($1,900,000 - $500,000) = 19%

Zale: ($850,000 - $200,000) / ($2,700,000 - $1,000,000) = 38%

Zale's retained cash flow (using CFO as operating cash) relative to its net debt level is greater than York. This, too, suggests a lower credit quality for York relative to Zale.

Overall, the picture is mixed. Relative to Zale, York is a more profitable company, suggesting a higher credit rating—though it relies more on debt in its capital structure, which suggests a lower credit rating. Both companies report similarly strong coverage ratios.

LOS 62.c: Describe the seniority rankings of debt, secured versus unsecured debt and the priority of claims in bankruptcy, and their impact on credit ratings.

Each different type of bond of a particular issuer is ranked according to a **priority of claims** in the event of a default. A bond's position in the priority of claims to the issuer's assets and cash flows is referred to as its **seniority ranking**.

Debt can be either **secured debt** or **unsecured debt**. Secured debt is backed by collateral, which can be sold to recover funds for bond investors in the event of default by the issuer. Unsecured debt represents a general claim to the issuer's assets and cash flows. Secured debt has a higher priority of claims than unsecured debt.

Secured debt can be further distinguished as *first lien* (where a specific asset is pledged) or *first mortgage* (where a specific property is pledged), *senior secured (second lien)*, or *junior secured* debt. Unsecured debt is further divided into *senior*, *junior*, and *subordinated* gradations. The highest rank of unsecured debt is senior unsecured. Subordinated debt ranks below other unsecured debt.

The general seniority rankings for debt repayment priority are the following:

1. First lien/mortgage
2. Senior secured (second lien)
3. Junior secured
4. Senior unsecured
5. Senior subordinated
6. Subordinated
7. Junior subordinated

All debt within the same category is said to rank **pari passu**, or have the same priority of claims. All senior secured debtholders, for example, are treated alike in a corporate bankruptcy. Any senior secured claims that are not met by the value of collateral upon which they are secured will automatically rank pari passu with senior unsecured claims in the bankruptcy process.

Recovery rates are the highest for debt with the highest priority of claims and decrease with each lower rank of seniority. The lower the seniority ranking of a bond, the higher its credit risk.

In a default or reorganization, senior lenders have claims on the assets before junior lenders and equity holders. A strict priority of claims, however, is not always applied in practice. Although in theory the priority of claims is absolute, in many cases, lower-priority debtholders (and even equity investors) may get paid even if senior debtholders are not paid in full in order to accelerate the bankruptcy process. Bankruptcies can be costly and take a long time to settle, during which the value of a company's assets could deteriorate due to loss of customers and key employees, while legal expenses mount.

Credit rating agencies rate both the issuer (i.e., the company issuing the bonds) and the debt issues, or the bonds themselves. Issuer credit ratings are called **corporate family ratings (CFRs)**, and are typically based on their senior unsecured debt, while issue-specific ratings are called **corporate credit ratings (CCRs)**.

The ratings of a firm's individual bonds can differ from its corporate (issuer) rating. The seniority and covenants (including collateral pledged) of an individual bond issue are the primary determinants of differences between an issuer's rating and the ratings of its individual bond issues. The assignment of individual issue ratings that are higher or lower than that of the issuer is referred to as **notching**.

Another example of a factor that rating agencies consider when notching an issue credit rating is **structural subordination**. In a holding company structure, both the parent company and the subsidiaries may have outstanding debt. A subsidiary's debt covenants may restrict the transfer of cash or assets "upstream" to the parent company before the subsidiary's debt is serviced. In such a case, even though the parent company's bonds are not junior to the subsidiary's bonds, the subsidiary's bonds have a priority claim to the subsidiary's cash flows. Thus, the parent company's bonds are effectively subordinated to the subsidiary's bonds with respect to the subsidiary's cash flows.

Notching is less common for highly rated issuers than for lower-rated issuers. For firms with high overall credit ratings, differences in expected recovery rates among a firm's individual bonds are less important, so their bonds might not be notched at all. For firms with higher probabilities of default (lower ratings), differences in expected recovery rates among a firm's bonds are more significant. For this reason, notching is more likely for issues with lower creditworthiness in general.

Reading 62: Key Concepts

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LOS 62.a

Qualitative factors that imply a corporate issuer has high creditworthiness include having a stable business model, being in an industry with low levels of competition, having low business risk, and having adequate corporate governance.

Quantitative factors that imply a corporate issuer has a high creditworthiness include strong profitability and recurring revenues, low leverage, high coverage ratios, and high levels of liquidity.

LOS 62.b

Credit analysts use profitability, leverage, and coverage ratios to assess debt issuers' capacity:

- Profitability can be measured by EBIT margin (EBIT / revenue).
- Coverage can be measured by EBIT to interest expense.
- Leverage ratios include debt to EBITDA and retained cash flow to net debt.

LOS 62.c

Corporate debt is ranked by seniority or priority of claims. Secured debt is a direct claim on specific firm assets and has priority over unsecured debt. Secured or unsecured debt may be further ranked as senior or subordinated. Priority of claims may be summarized as follows:

1. First lien/mortgage
2. Senior secured (second lien)
3. Junior secured
4. Senior unsecured
5. Senior subordinated
6. Subordinated
7. Junior subordinated

Issuer credit ratings, or corporate family ratings (CFRs), reflect a debt issuer's overall creditworthiness and typically apply to a firm's senior unsecured debt.

Issue credit ratings, or corporate credit ratings, reflect the credit risk of a specific debt issue. Notching refers to adjusting an issue credit rating upward or downward from the issuer credit rating.

For a specific debt issue, secured collateral implies lower credit risk compared to unsecured debt, and higher seniority implies lower credit risk compared to lower seniority.